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Review article

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Epidemiology of cardiovascular–kidney–metabolic syndrome

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Abstract

Cardiovascular–kidney–metabolic (CKM) syndrome, defined by the American Heart Association in 2023, describes the interconnected risks of metabolic dysfunction, chronic kidney disease and cardiovascular disease. By unifying these conditions, CKM highlights shared pathophysiology, overlapping risk factors and common therapeutic approaches. The staging framework, from ideal CKM health (stage 0) to clinical cardiovascular disease (stage 4), underscores the importance of prevention and captures the escalating risk of morbidity and mortality with advancing stage. Epidemiological data demonstrate increasing global prevalence and that most adults already meet criteria for stage ≥1 CKM syndrome, with stage 2 (defined by metabolic risk factors or the presence of kidney disease) being the most common. The burden of CKM syndrome is characterized by demographic, socio-economic and geographic disparities, with men and older adults most affected. Established determinants of CKM health include genetics, diet, smoking and physical inactivity, as well as social and structural factors; emerging environmental risk factors, such as climate-related stressors, are increasingly investigated but have a less established evidence base. Lifestyle interventions remain foundational and modern therapeutic classes — including glucagon-like peptide-1 receptor agonists, sodium–glucose co-transporter-2 inhibitors and non-steroidal mineralocorticoid receptor antagonists — offer multisystem benefits, although cost, access and variable uptake remain important barriers. Here, we examine current CKM epidemiological evidence and highlight priorities for prevention, equity and future research.

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Key points

- Cardiovascular–kidney–metabolic (CKM) syndrome was formally defined by the American Heart Association in 2023 to unify cardiovascular disease, chronic kidney disease and metabolic risk factors into a single multisystem construct.
- The staging system (stages 0–4) highlights opportunities for prevention and early detection, with advancing stages being associated with increasing cardiovascular and all-cause mortality.
- Given global trends in obesity, most adults worldwide already meet criteria for stage ≥1 CKM syndrome, with stage 2 (defined by the presence of metabolic risk factors or chronic kidney disease) being the most prevalent. Advanced stages disproportionately affect men, older adults, and socio-economically disadvantaged and rural populations globally.
- CKM health is shaped by genetic, dietary, social, structural, behavioural and environmental determinants; emerging risk factors include climate-related stressors.
- Modern therapeutics, including glucagon-like peptide-1 receptor agonists, sodium–glucose co-transporter 2 inhibitors and non-steroidal mineralocorticoid receptor antagonists, show multisystem benefits. but their real-world impact is constrained by cost and access barriers, leading to variable uptake.
- Although the CKM framework promotes integrated care and risk prediction, concerns including the classification of nearly all adults as having disease warrant scrutiny. Key research priorities include global epidemiological characterization and longitudinal tracking of stage transitions, as well as addressing current inequities in the uptake of proven therapies.

Introduction

Cardiovascular, kidney and metabolic diseases are leading causes of morbidity and mortality worldwide, with substantial and long-recognized clinical overlap[1]. In October 2023, the American Heart Association (AHA) formally defined cardiovascular–kidney–metabolic (CKM) syndrome with simultaneous publication of a Presidential Advisory and Scientific Statement[2,3]. Numerous studies have since evaluated the CKM concept, demonstrating its growing uptake in epidemiological research and clinical practice[4–6].

Multidisciplinary experts convened by the AHA described CKM syndrome as "a systemic disorder characterized by pathophysiological interactions among metabolic risk factors, chronic kidney disease (CKD) and the cardiovascular system, leading to multiorgan dysfunction and a high rate of adverse cardiovascular outcomes." The formalization of CKM syndrome followed decades of recognition of shared risk factors, multidirectional interactions and common pathophysiology underlying cardiovascular disease (CVD), CKD and metabolic risk factors (for example, hypertension, hypertriglyceridaemia and type 2 diabetes (T2D)). The development of therapies with pleiotropic benefits across CKM health — including glucagon-like peptide-1 receptor agonists (GLP-1RAs) and sodium–glucose co-transporter 2 (SGLT2) inhibitors — further motivated the AHA definition. By unifying these multisystem manifestations, the AHA aimed to improve identification of biological and social drivers of CKM health, enable earlier detection of patients most in need of intervention and guide integrated prevention and treatment strategies. The CKM construct also provides a single framework for research, health system planning and policy.

CKM syndrome is conceptualized as a continuum of five stages[2], from stage 0 to 4 (Fig. 1). Stage 0 denotes ideal CKM health, characterized by healthy body weight and adiposity, normoglycaemia, normotension, a normal lipid profile and the absence of CKD and CVD. Stage 1 reflects early manifestations of excessive or dysfunctional adipose tissue, including overweight (BMI ≥ 25 kg/m^2), obesity (BMI ≥ 30 kg/m^2) or impaired glucose tolerance. Thresholds for excess adiposity differ by ancestry and sex, with lower BMI cutoffs for individuals of Asian ancestry (≥23 kg/m^2) and sex-specific waist circumference cutoffs. Stage 2 is defined by ≥1 metabolic risk factors — hypertriglyceridaemia (≥135 mg/dl), hypertension (stages 1 and 2) and T2D — and/or moderate- to high-risk CKD, defined by Kidney Disease Improving Global Outcomes (KDIGO) as a combination of albuminuria and/or reduced glomerular filtration rate (GFR). Stage 3 is characterized by asymptomatic or subclinical CVD in individuals who otherwise meet stage 1 or stage 2 criteria. Diagnosis might be based on elevated coronary artery calcium, abnormal cardiac biomarkers, imaging evidence of CVD or the presence of subclinical CVD risk equivalents, including G4 or G5 stage CKD (that is, GFR <30 ml/min/1.73m^2) or high predicted CVD risk (for example, 10-year Predicting Risk of CVD EVENTs (PREVENT) risk ≥20%). Finally, individuals with CKD, metabolic risk factors or excess adiposity who develop symptomatic CVD are classified as having stage 4 CKM syndrome[2].

In this Review, we summarize current evidence on prevalence, incidence and temporal trends in CKM health, as well as examining genetic, social and environmental determinants of CKM outcomes. We also assess the real-world impact of emerging therapeutics and discuss the clinical implications, strengths and limitations of the CKM paradigm. Throughout, we highlight future directions for CKM-focused interventions and research.

Prevalence, incidence and trends in CKM syndrome

Estimating the burden of cardiovascular–kidney–metabolic syndrome in populations

Treating comorbid cardiovascular, kidney and metabolic diseases within CKM syndrome is clinically rational. However, CKM syndrome is newly defined, and the field faces several challenges. Limited epidemiological data mean that the burden of CKM syndrome must be partly extrapolated from data on its individual component diseases. Changes in screening guidelines and variable adherence further complicate the interpretation of secular trends. Data on syndrome progression and stage transitions are scarce, as are studies of low-income global populations. Despite these gaps, high-quality analyses, most notably from nationally representative surveys such as the National Health and Nutrition Examination Survey (NHANES) have already enriched the understanding of the impact of CKM syndrome at a population level.

Prevalence of cardiovascular–kidney–metabolic syndrome in the USA

Initial studies have begun to characterize the epidemiology of CKM syndrome in the contemporary US population (Fig. 2). In a 2024 study of 10,762 adults from NHANES (2011–2020)[4], 10.6% met stage 0 criteria, 25.9% stage 1, 49.0% stage 2, 5.4% stage 3 and 9.2% stage 4.

Adults aged 45–65 years had significantly more advanced CKM stages than those aged 20–44, as did men compared with women, and Black adults compared with white adults. Prevalence remained stable during the study period. Another study of a similar NHANES population (2011–2018), reported nearly identical stage distributions and demographic disparities[7]. Outside NHANES, data from the Behavioral Risk Factor Surveillance System suggested that 79.5% of adults in the USA had stages 1–4 CKM syndrome, with stage 1 being the most common (41.6%) and stage 4 affecting 6.7%[8]. Finally, an analysis of de-identified electronic health records and claims data from the OptumLabs Data Warehouse (OLDW) database estimated that 8.7% of people in 2023 had ideal CKM health (stage 0), whereas stage 2 affected 48% and stage 4 affected 22% of the cohort (L.A.M. and M.E.G., unpublished work). The higher prevalence of stage 4 in the OLDW cohort compared with nationally representative surveys probably reflects the older age of the cohort (mean 56.9 years), inclusion of only health care-seeking individuals and more contemporary data.

These studies provide several key insights. Most individuals in the USA with overweight or obesity have additional metabolic risk factors or CKD, explaining the higher prevalence of stage 2 than stage 1. Although stage 2 CKM syndrome requires only one metabolic risk factor or CKD, multimorbidity is the norm at this stage. Among people with T2D, for example, <10% lacked additional CKM pathology and nearly 20% had CK[9]. Finally, stage 0 CKM syndrome and advanced CKM syndrome have similar prevalence.

Global prevalence and interregional variation

Efforts have been undertaken to estimate the burden of CKM syndrome globally, and the most comprehensive non-US data are from East Asia. In China, nationally representative surveys showed that CKM syndrome is extremely common in middle-aged and older adults, with a slightly lower prevalence than in the USA (83.7% versus 89.4%) and a lower prevalence of advanced stages (10.1% versus 14.6%)[10,11]. Stage 2 was the most common, and advanced CKM syndrome disproportionately affected rural residents, men and older adults. Similar rural–urban and sex disparities were observed in Korean adults, although the overall prevalence of CKM syndrome in Korea (74.8%) was lower than in China and the USA[12].

Data on CKM syndrome prevalence in Europe are sparser. One Italian primary care study reported an increase in CKM syndrome prevalence between 2017–2019 and 2020–2022, but its interpretation is limited by non-representative sampling[13]. UK Biobank data from adults aged 40–69 years between 2006 and 2010 revealed a predominance of stage 2 CKM syndrome. Only 12.7% of adults were free of CKM risk factors (stage 0) and men were more likely than women to have advanced comorbidities[14]. These findings resemble trends observed in the USA and East Asia.

Outside of the high-income regions of Asia, Europe and North America, data on CKM syndrome prevalence are extremely limited. A Global Burden of Disease 2021 analysis showed that most of the global population is affected by CKM conditions, with a rising burden driving increasing deaths and disability-adjusted life years (DALYs) in low-to-middle-income countries in Southeast Asia, Africa, the Western Pacific and Eastern Mediterranean[1]. International efforts to characterize stage distributions and disparities are paramount to improving CKM health equitably.

Secular trends in syndrome burden

In the USA, NHANES data showed that the proportion of adults at stage 0 (ideal health) declined by nearly 50% between 1988 and 2018.

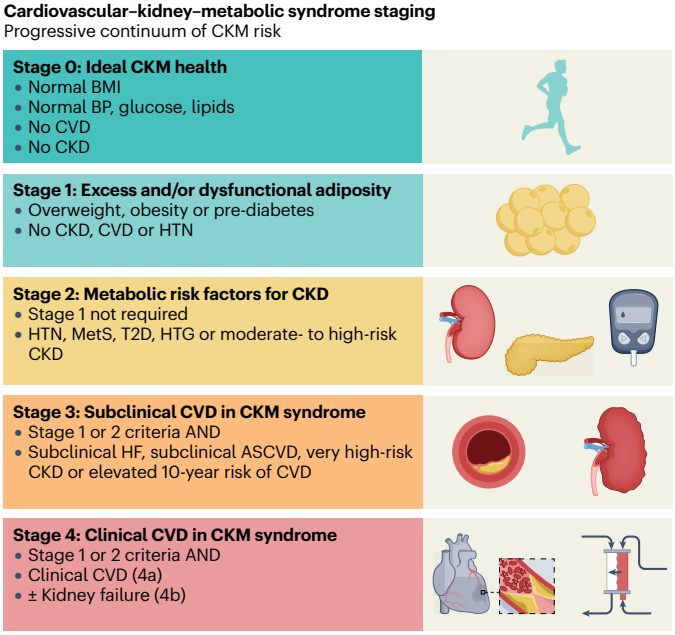


Fig. 1 | Cardiovascular–kidney–metabolic syndrome staging criteria. Diagnostic criteria for cardiovascular–kidney–metabolic (CKM) syndrome stages 0–4, according to the American Heart Association[2,3]. Stage 0 represents ideal CKM health. Stage 1 is defined by either overweight, obesity or insulin resistance in the absence of additional CKM pathology. Stage 2 is defined by type 2 diabetes, hypertension, hypertriglyceridaemia, metabolic syndrome or moderate- to high-risk chronic kidney disease (CKD) according to Kidney Disease Improving Global Outcomes criteria; meeting stage 1 criteria is not required. Stage 3 is defined by stage 1 or 2 criteria and subclinical cardiovascular disease, very high-risk CKD or ≥20% 10-year cardiovascular disease (CVD) risk. Stage 4 is defined by stage 1 or 2 criteria and symptomatic CVD with (stage 4b) or without kidney failure (stage 4a). ASCVD, atherosclerotic CVD; BP, blood pressure; HF, heart failure; HTG, hypertriglyceridaemia; HTN, hypertension; MetS, metabolic syndrome; T2D, type 2 diabetes.

Over the same period, more advanced CKM syndrome stages became increasingly prevalent, particularly among men, with stage 3 rising substantially and stage 2 remaining the most common in both sexes[15]. In China, the proportion of individuals free of CKM risk factors fell by approximately 30% during the 2010s[11]. These trends are consistent with the global rise in CKM syndrome components, including diabetes[16], obesity[17] and CKD[18].

There are multiple reasons for the global decline in CKM health. The global population is ageing, and age is a leading CKM risk factor. However, population ageing alone does not explain the decline in CKM health. Age-standardized DALY rates of stroke, peripheral artery disease and ischaemic heart disease fell between 1990 and 2021. By contrast, upstream drivers — particularly high BMI and insulin resistance — increased in an age-independent manner, suggesting that rising CKM syndrome burden is driven primarily by metabolic deterioration rather than demographic shifts[1]. The global rise in obesity is one of the defining public health trends of the past 50 years, with obesity rates rising from 3% to 11% among men and 6% to 15% among women between 1975 and 2015 (ref. 17).

These temporal epidemiological patterns have been described in an ‘obesity transition’ model[17]. In countries in the first stage

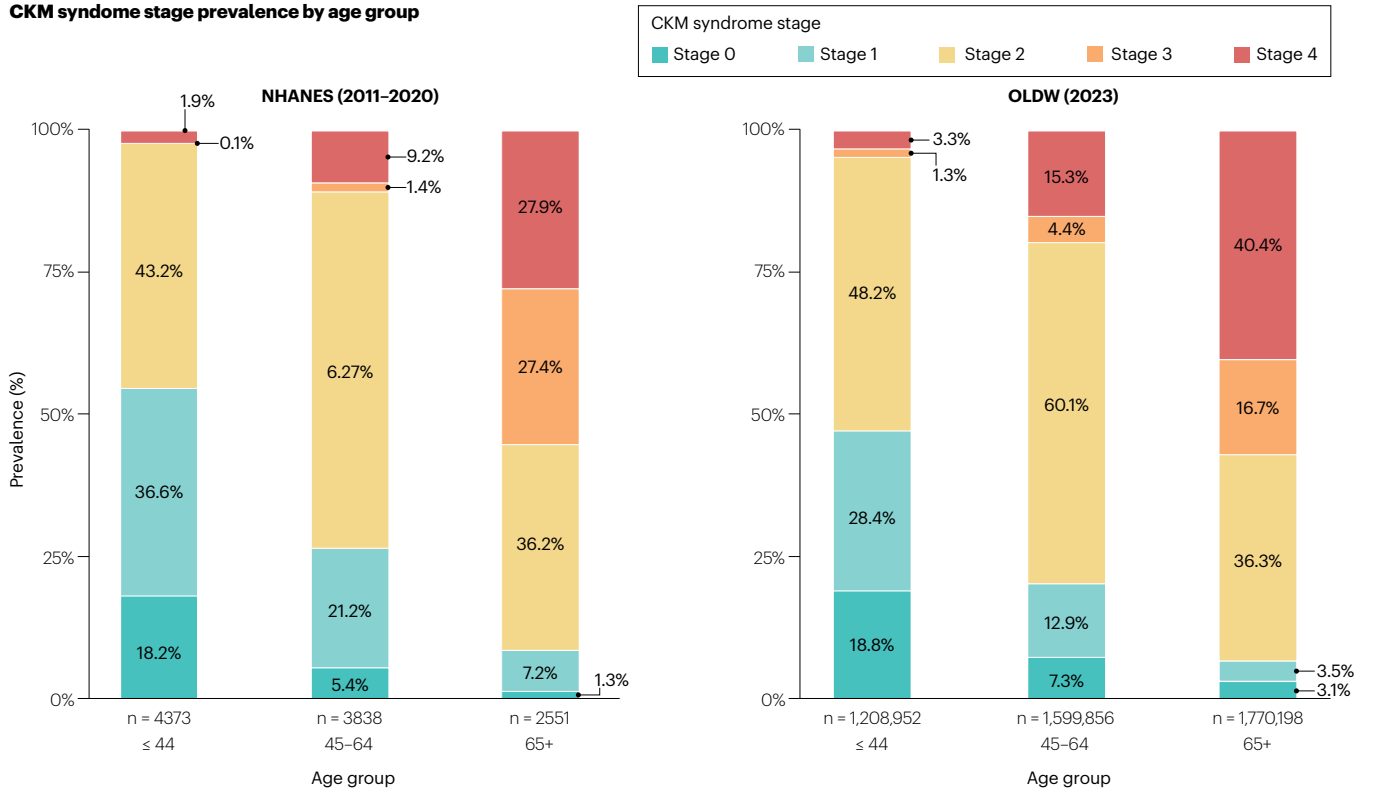


Fig. 2 | Cardiovascular–kidney–metabolic syndrome stage prevalence in adults in the USA by age group. The graph represents age-stratified cardiovascular–kidney–metabolic (CKM) syndrome prevalence in National Health and Nutrition Examination Survey (NHANES) (2011–2020)[4] and OptumLabs Data Warehouse (OLDW) (2023). Stacked bars display the distribution of CKM syndrome stages (Fig. 1) across age strata. The OLDW analysis included all individuals with a BMI measurement in 2023 and at least 1 year of continuous enrolment prior to the index date. Syndrome stages were defined by a combination of structured clinical data, natural language processing data extracted from clinical notes, International Classification of Diseases codes and medication use. NHANES sample sizes are unweighted.

(for example, many sub-Saharan African nations), obesity affects a small fraction of the population, predominantly women of high socio-economic status. In the second stage, obesity becomes more widespread among adults, sex and socio-economic differences narrow and increases appear in children (a pattern seen in many contemporary East Asian populations). Obesity then begins to disproportionately affect lower socio-economic groups, plateauing in women and children from high-income groups. The USA exemplifies this phase of the obesity transition. Stage 4, a population-level decline in obesity, has yet to be convincingly observed.

The global increase in BMI has incompletely understood causes[19]. Obesity is strongly associated with the 'westernization' of lifestyles[20], reliance on ultra-processed, calorie-dense foods[21,22] and built-environment factors such as poor access to nutritious food[23]. Importantly, although genetic predisposition might influence individual vulnerability to obesity, food policy and the local food environment[23,24] are likely to affect obesity rates within countries and regions[25].

As emphasized by the AHA[2], excess and dysfunctional adiposity are central to CKM syndrome pathogenesis. Thus, although the causes of the obesity transition remain incompletely understood, targeting the global rise of obesity will be crucial to tackling the concomitant increase in CKM syndrome prevalence.

Epidemiology of cardiovascular–kidney–metabolic syndrome sequelae

The adverse sequelae of CKM syndrome are closely linked to stage (Fig. 3). Globally, ischaemic stroke and heart disease are the leading CKM-related causes of DALYs lost and disproportionately affect individuals in their seventh and eighth decades in middle sociodemographic index regions.[1] The risk of these complications increases progressively with CKM stage. In a large Taiwanese cohort, the risk of CVD and stroke mortality tripled at CKM syndrome stage 2 and increased further at stages 3 and 4 (ref. 26). A US-based NHANES analysis[5] revealed a similar stepwise increase, with CVD mortality nearly tripling from stages 0–1 to stage 4. This study also showed that all-cause and non-CVD mortality approximately doubled from ideal CKM health to stage 4. Studies in British and Chinese cohorts yielded similar findings[27,28]. Together, these results highlight stage 1 as a critical window for intervention to prevent progression of CKM syndrome to late-stage disease.

The latest evidence suggests that sociodemographic factors might modify the relationship between CKM stage and mortality. Multiple studies show that at every stage, women have excess mortality risk compared with men[5,15]. Social and environmental determinants of health — including fine particulate air pollution, poor access to nutritious foods[29] and low dietary anti-oxidant intake[30] — are

also associated with increased age-adjusted mortality. Additionally, higher age has not only been associated with more advanced CKM syndrome[4,7,8,10–14] but also with increased risk of mortality conferred by advanced stages[31].

Despite these insights, several gaps remain. Most studies have focused on mortality, whereas the impact on other important endpoints is less clear. Early-stage CKM components such as high BMI and diabetes are major contributors to years lived with disability globally[1]. More research is needed to characterize the full spectrum of CKM-related outcomes, including morbidity, disability and quality of life. Additionally, the natural history of CKM syndrome remains poorly characterized. Data on time spent in each stage, rates and predictors of upstaging and downstaging, and clinical consequences of these transitions are lacking. Addressing these gaps will be essential to combat this condition.

Determinants of cardiovascular–kidney–metabolic health

Shared risk factors underlie the overlap of cardiovascular, kidney and metabolic disease. Biological, social and environmental determinants promote excess adipose tissue, triggering molecular cascades that drive metabolic dysfunction and CKD. Dysfunction in one CKM system exacerbates the others, provoking syndromic progression that can lead to symptomatic CVD, kidney failure, hospitalizations and death (Fig. 4). Beyond their effects on adiposity, many risk factors directly promote downstream CKM complications. Dietary factors, such as high intake of ultra-processed foods (UPFs) and excess caloric load, drive CVD risk by increasing atherogenic lipoproteins[32]. Social stressors, including limited access to preventative care, worsen CKM health[33,34]. Environmental exposures such as cigarette smoking might exacerbate vascular injury and kidney dysfunction[35,36]. These determinants shape

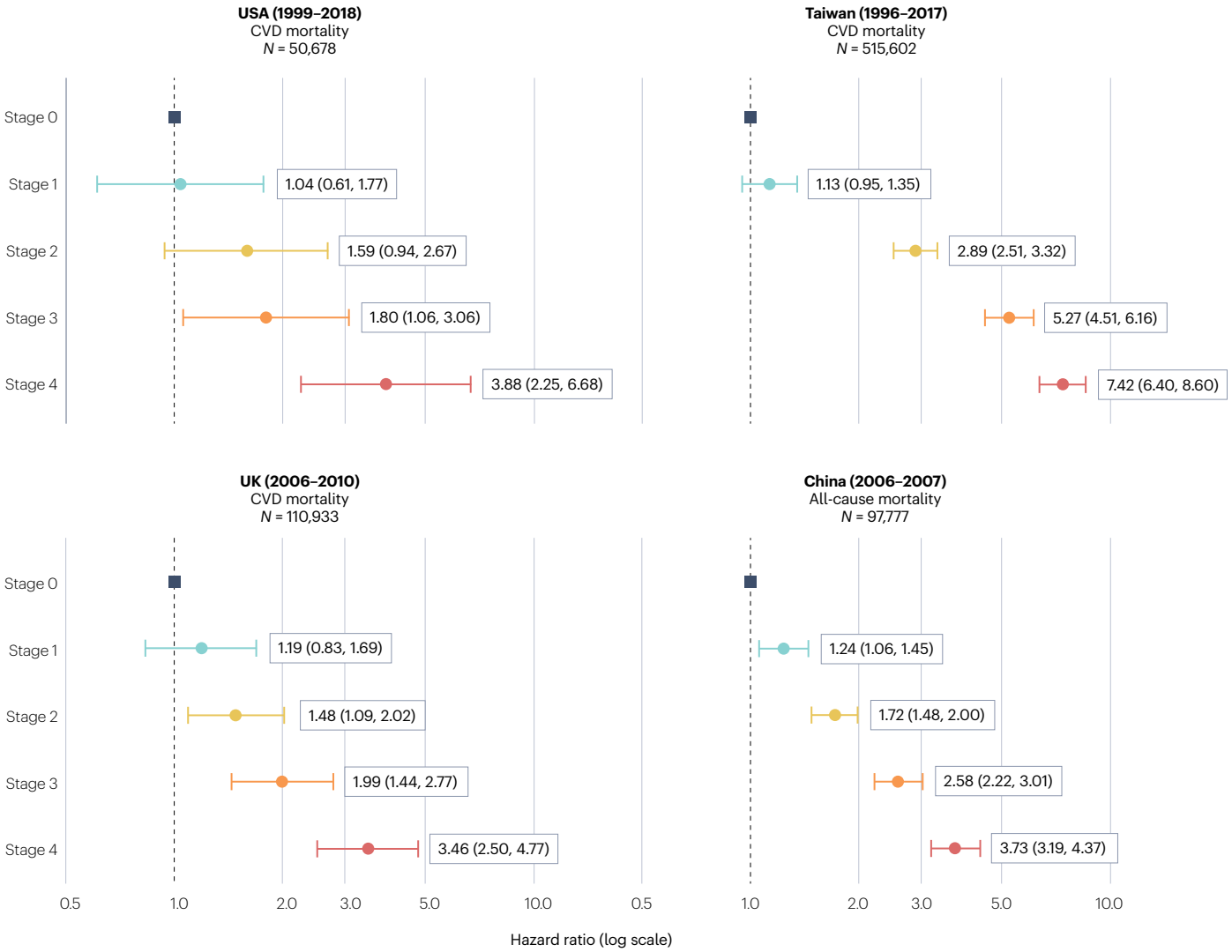


Fig. 3 | Cardiovascular–kidney–metabolic stage-specific mortality risk across global cohorts. Forest plots of cardiovascular–kidney–metabolic (CKM) syndrome stage-specific mortality risk display US (1999–2018, cardiovascular disease (CVD) mortality)[5], Taiwanese (1996–2017, CVD mortality)[26], UK (2006–2010, CVD mortality)[27] and Chinese (2006–2007, all-cause mortality)[28] cohorts. Panels note study sample size and end point. Adjustment for covariates and definitions of CKM stages differ by study.

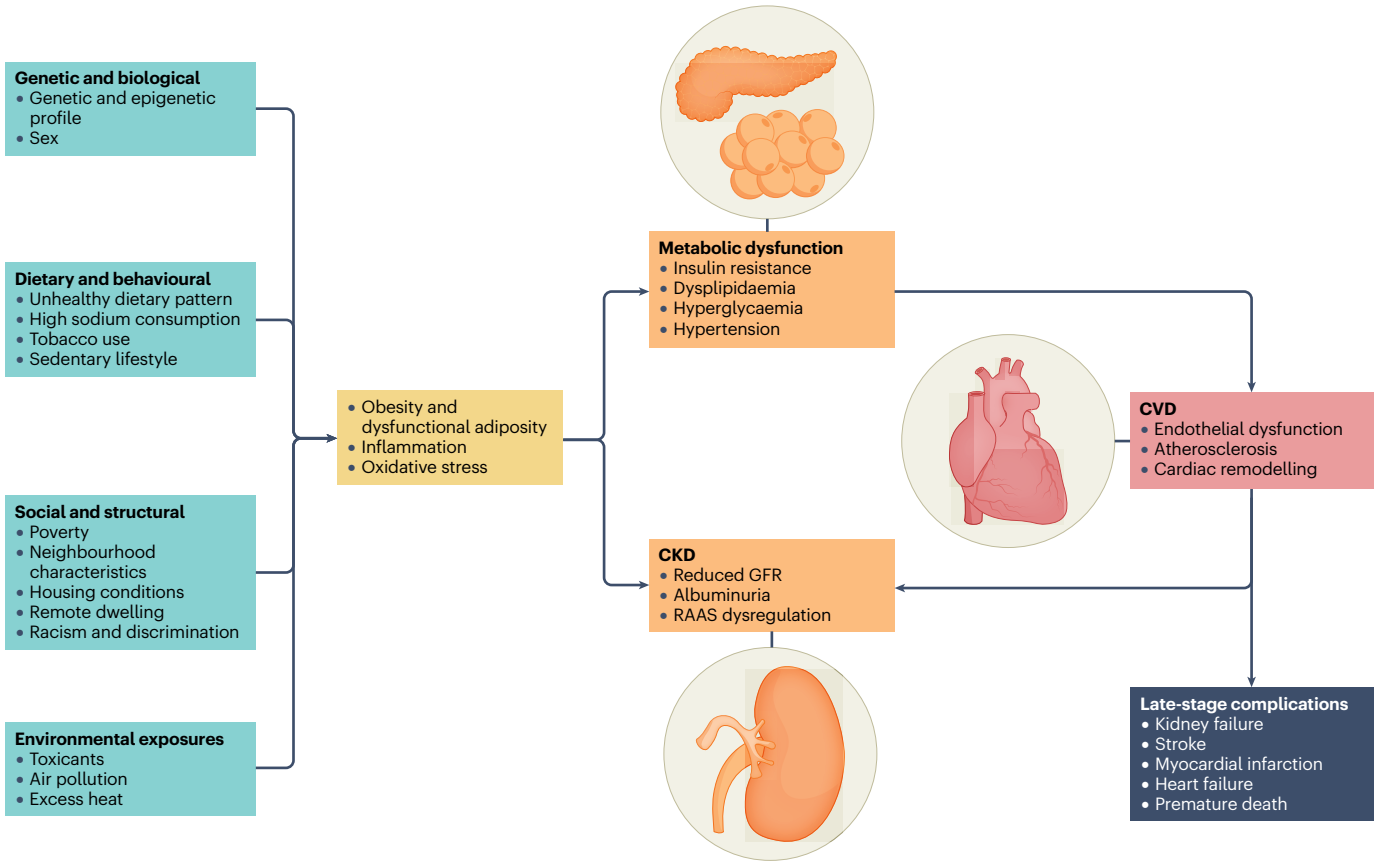


Fig. 4 | The pathophysiology of cardiovascular–kidney–metabolic syndrome. Upstream determinants of cardiovascular–kidney–metabolic (CKM) syndrome include genetic and biological factors (family history and polygenic risk, sex), dietary and behavioural exposures (excess calories, high sodium, tobacco, physical inactivity), social and structural conditions (poverty, built environment, rural residence) and environmental stressors (climate change, air pollution, microplastics). Collectively, these determinants interact and promote excess or dysfunctional adiposity, adipose inflammation and oxidative stress, which trigger metabolic dysfunction characterized by insulin resistance, dyslipidaemia, hypertension and inflammation. In addition to metabolic dysfunction, these determinants are implicated in the development of chronic kidney disease (CKD), characterized by glomerular hyperfiltration, albuminuria and dysfunction of the renin–angiotensin–aldosterone system (RAAS). These processes accelerate cardiovascular disease (CVD) through endothelial dysfunction, atherogenesis and cardiac remodelling. CKD, CVD and metabolic syndrome therefore form a feedforward loop maintained by processes such as maladaptive neurohormonal signalling, inflammatory cytokine pathways, altered adipokine profiles and cardiac pressure overload. Over time, this network of disease-promoting alterations can lead to late-stage complications including kidney failure, stroke, myocardial infarction, heart failure and premature death.

the incidence, progression, distribution and secular trends of CKM syndrome. Although much of the research on CKM determinants examines individual components, a growing body of evidence — particularly on social and structural determinants — reflects the integrated AHA framework. Together, existing studies highlight opportunities for risk prediction and prevention, while underscoring areas for future research.

Genetic epidemiology

Genomic factors are strongly implicated in the pathogenesis of CKM diseases. Heritability estimates for CKM traits range from 20% to 60%, with higher values for BMI and lipid levels, and lower values for kidney function, kidney damage and heart failure[37–39]. Although substantial genetic overlap between CKM components exists, the genetics of CKM syndrome have not been studied as a unified entity; rather, existing research examines individual traits. A 2022 genome-wide association study (GWAS) from the CARDIoGRAMplusC4D4 consortium compared 181,522 individuals with coronary artery disease (CAD) to 1,165,690 control individuals and identified 241 associations at 198 loci. These findings implicated early developmental processes, cell-cycle signalling, and vascular cell migration and proliferation in CAD pathogenesis[40]. A large GWAS from the CKD Genetics Consortium and UK Biobank identified 634 independent signals, explaining 9.8% of estimated GFR (eGFR) variance — a smaller proportion than that observed for other CKM traits, consistent with the lower heritability of kidney function[41]. Other GWAS and whole-exome studies have described CKM-associated phenotypes such as diabetes[42], obesity[43] and metabolic dysfunction-associated steatotic liver disease (MASLD)[44].

Genetic epidemiological approaches have elucidated biological subtypes and overlap of CKM components. A large multi-ancestry GWAS of T2D clustered variants into mechanistic groups with distinct cardiometabolic profiles — including clusters linked to metabolic

syndrome, obesity and body fat — highlighting biological heterogeneity within metabolic syndrome, although kidney traits were not examined[42]. Emerging work has begun to characterize genetic and molecular overlap across CKM components. A Swedish twin study estimated that approximately half of the phenotypic correlation between metabolic syndrome and CKD is attributable to shared genetic factors[45]. Trans-ethnic Mendelian randomization analyses identified causal effects of BMI, hypertension, T2D and lipid traits on CKD risk, with notable differences between European and East Asian populations[46]. Later kidney proteogenomic analyses identified shared molecular pathways linking kidney function to cardiovascular and metabolic traits, highlighting potential therapeutic targets for CKM diseases[47].

A persistent challenge in genetic epidemiology is the historical focus on European ancestry populations. Under-representation of non-European individuals limits generalizability and risks worsening health inequities[48]. Recent multi-ancestry GWAS show the value of diverse cohorts as they identify more variant-trait associations[42]. The discovery of *APOL1* as a driver of kidney disease in populations of West African ancestry exemplifies the importance of studying diverse populations[49,50]. Common *APOL1* variants, which occur nearly exclusively in individuals of African ancestry, contribute to high kidney failure rates and might modestly increase CVD risk[51,52], which has prompted therapeutic targeting in ongoing trials[49,50,53].

Dietary risk factors

Even among individuals with a high genetic risk of CKM syndrome, environmental factors remain major drivers of CKM health[54]. Although the genetic profiles of populations have changed minimally over recent decades, CKM health has markedly deteriorated. A key environmental determinant is diet, with one in five global deaths associated with poor diet in 2017, the largest estimated contributions being from cardiovascular disease and, to a lesser extent, T2D[55]. Excess sodium and insufficient consumption of whole grains, fruits, vegetables and seeds are leading dietary risk factors for CVD-associated deaths and DALYs. Processed meats and sugar-sweetened beverages are smaller contributors globally but have an outsized role in North America[55].

Beyond specific foods, overall dietary patterns might shape CKM health. Epidemiological data show that in high-income nations, UPF consumption is particularly impactful. UPFs are made predominantly from industrial substances, contain little or no whole foods (that is, edible parts of plants and animals) and are engineered to prolong shelf-life, palatability, convenience and encourage excess caloric intake. UPFs are typically high in sodium, added sugars and saturated fat, and low in fibre and protein, a profile associated with hypertension, insulin resistance and dyslipidaemia[56]. In a large Spanish cohort study, each additional daily serving of UPFs was associated with an 18% increase in all-cause mortality[57]. UPFs are also associated with higher risks of overweight and obesity, hypertension, T2D, CKD and CVD mortality in international cohorts[58,59].

By contrast, dietary patterns such as the Mediterranean, Dietary Approaches to Stop Hypertension (DASH) and plant-based diets have protective associations with incident and progressive CKM disease, and all-cause mortality. The Mediterranean diet is high in fruits and vegetables, legumes and whole grains, fish and olive oil, and is associated with lower risks of CVD, T2D and metabolic disease[60]. The DASH diet reduces blood pressure, with greater effects in the setting of reduced sodium intake[61,62]. Plant-based diets lower low-density lipoprotein cholesterol, glycated haemoglobin (HbA_{1c}) and body weight in individuals at a high risk of CVD[63], although benefits depend on diet composition and quality[64].

Although establishing causality in nutritional science is challenging owing to high research costs and limited funding, as well as the difficulty in isolating distinct dietary components and addressing potential confounders such as levels of physical activity and body composition, the major public health implications of diet quality underscore the need for sustained investment and rigorous research to improve CKM health.

Social and structural determinants

There are notable socio-economic, racial and rural–urban disparities in CKM health. Many disparities are shaped by social and structural factors and the built environment. Social determinants include individual-level factors such as income, education and employment; structural determinants refer to policies and systems shaping resource distribution; and the built environment encompasses physical infrastructure, including housing, transportation, and access to green space and nutritious food. Social risk profile — including employment status, income level, educational attainment, home ownership and marriage status — is strongly associated with CKM stage, independently of demographic and lifestyle factors[65]. Social risk is also associated with CKM-related mortality: individuals with low incomes, rural residence and low educational attainment die at higher rates than their higher income counterparts[29].

Social risks are reflected in and perpetuated by neighbourhood conditions shaped by past and present policies. In the USA, 'redlining' exemplifies the interplay between policy, economics, sociology and CKM health. This 1930s policy graded neighbourhoods partly by racial and ethnic composition, leading to long-term underinvestment in minoritized communities. A 2025 scoping review showed that redlined neighbourhoods have a higher prevalence of and worse outcomes for atherosclerotic CVD (ASCVD), heart failure (HF), CKD and other cardiometabolic risk factors[66]. Analogous policy-level determinants exist beyond the USA: in Indigenous populations in Australia and Canada, for example, colonial policy legacies, structural racism and geographic and/or service inequities are linked to higher cardiometabolic and kidney disease burden[67–70].

There are multiple links between socio-economic status, the built environment and CKM health. Exposure to air pollution, particularly particulate matter ≤2.5 μm (PM2.5), disproportionately affects low-income neighbourhoods in the USA[71] and low-income countries globally, and is estimated to account for ~20% of global T2D cases[72] and higher CKD rates[73]. Food insecurity is more common in historically neglected US neighbourhoods and is associated with disparities in CKM health[74]. Urban design also matters: cities supportive of physical activity (for example, through park access, transit and residential density, walkability and safety) are associated with greater physical activity and lower obesity[75]. Lower-income and majority–minority neighbourhoods often lack access to green space and report lower perceived safety, even when theoretically walkable[76,77].

Rural residence is also a risk factor for poor CKM health globally. In China, rural residents have a nearly 2-year lower life expectancy than urban residents. More than 50% of this excess mortality has been attributed to circulatory diseases[78]. Similar rural–urban disparities in CKM disease burden and outcomes exist in the USA[79], South Asia[80] and Australia[81]. Although causes vary across settings, policy failures, poor access to health care and broader social risks linked to rural residence all contribute globally.

Toxic exposures and behavioural determinants

Several toxic exposures and behavioural factors shape CKM health. Large meta-analyses show that tobacco smoking increases incident CVD risk in a dose–response manner, with one cigarette per day associated with ~50% higher CVD risk and a greater relative risk increase among women[82]. Meta-analyses show that smoking increases risk of CKD, kidney failure and T2D by similar magnitudes, with risk regression observed only decades post-cessation[83,84].

Exposure to toxic metals, including cadmium and lead, is another contributor to CKM disease[85–87]. Heavy metal exposure is robustly associated with increased risk of CKD, CVD and diabetes in epidemiological studies, with proposed mechanisms including oxidative stress, depletion of endogenous anti-oxidants and endothelial damage[88–90].

A sedentary lifestyle also contributes to CKM component conditions. Higher daily sedentary time is associated with increased metabolic syndrome and CKD risk[91,92], and physical inactivity accounted for 6% of global CAD burden and 7% of global T2D burden in a 2012 study[93]. Sedentary time is also associated with all-cause and CVD-specific mortality in a dose–response manner[94]. These associations persist after adjusting for moderate-to-vigorous physical activity or cardiorespiratory fitness[95].

Emerging risk factors

Several novel risk factors have been investigated in relation to CKM health, although evidence remains less established than for traditional determinants. A systematic review demonstrated that high temperature, extreme weather events, such as hurricanes, and ground-level ozone[96] were associated with increased CVD morbidity and mortality risk. Extreme heat exposure and hurricanes were also associated with mortality among people with advanced CKD[97,98]. These climate-related stressors might disproportionately affect disadvantaged populations as climate change intensifies their frequency and severity[96,99].

Mechanistic animal studies suggest that microplastics and nanoplastics accumulate in kidney tissue and disrupt glucose metabolism and lipid homeostasis[100,101]. Human evidence remains limited, although one study showed that carotid artery plaques containing micro- and nanoplastics were associated with >4-fold increased cardiovascular event risk[102]. The clinical significance and major exposure sources of micro- and nanoplastics remain to be established.

Vaping and e-cigarettes have also been linked to negative long-term CKM effects, but this evidence comes exclusively from animal models[103]. Given widespread vaping prevalence, monitoring associations with disease in humans is critical and will rely on standardized clinical data quantifying e-cigarette exposure.

Real-world impact of novel therapeutics for cardiovascular–kidney–metabolic syndrome

The treatment of CKM syndrome has advanced greatly over the past decade (Fig. 5). Lifestyle interventions — including diet, exercise and smoking cessation — remain foundational, particularly for prevention and early-stage disease, and bariatric surgery offers durable benefits in severe obesity. Pharmacologically, landmark randomized controlled trials (RCTs) demonstrated that several drug classes have pleiotropic benefits across CKM syndrome components (Fig. 6), upending a paradigm in which CKM risk factors were managed with drugs targeting narrow, organ-specific indications. GLP-1RAs, SGLT2 inhibitors and non-steroidal mineralocorticoid receptor antagonists (nsMRAs) are the most notable examples, with additional therapeutic targets on the horizon. However, the real-world impact of these advances remains constrained by high costs, suboptimal risk-targeting and disparities in access — key challenges for patients, clinicians, researchers and policymakers.

Non-pharmacological interventions

Lifestyle interventions are central to CKM syndrome management, particularly for early-stage prevention. Meta-analyses of RCTs support aerobic and resistance-based exercise to reduce systolic and diastolic blood pressure, insulin resistance and intrahepatic fat content[104]; lower high-density lipoprotein cholesterol (HDL-C) and triglycerides[105]; and modestly reduce BMI and visceral fat[106]. These effects are most pronounced in moderate-to-high intensity supervised exercise programmes. Exercise might also slow eGFR decline in older adults and reduce hospitalizations and mortality in HF with reduced ejection fraction[107,108]. Importantly, exercise is relatively safe and low-cost. Dietary interventions, including the Mediterranean and DASH diets discussed previously, offer complementary benefits and are recommended across CKM stages.

For individuals with severe obesity who do not achieve adequate weight loss with lifestyle modification or pharmacotherapy, bariatric surgery is a highly effective non-pharmacological intervention. Long-term studies demonstrate durable weight loss of 20–30% of total body weight, along with improvements in glycaemic control, MASLD, blood pressure, kidney function and long-term cardiovascular risk[109,110]. Adverse effects include early and late surgical complications, micronutrient and vitamin deficiencies, and urolithiasis[111]. Despite a favourable risk–benefit profile, bariatric surgery is markedly underutilized. In the USA, fewer than 1% of eligible patients undergo the procedure annually, with lower rates among racial and ethnic minorities, rural populations and low income groups[112,113]; European and Asian data suggest similarly low utilization[114,115]. Barriers include insurance coverage, insufficient referral from primary care, patient concerns about surgical risk and a shortage of bariatric surgery centres in underserved areas[114,116]. For most patients not undergoing surgery, pharmacotherapy remains the primary approach to CKM risk reduction beyond lifestyle modification.

Glucagon-like peptide-1 receptor agonists

GLP-1RAs, including newer dual incretin agonists, have demonstrated benefits in obesity management, cardiovascular risk reduction and CKD progression in people with and without T2D and thus have the potential to halt and downstage CKM syndrome. High-dose semaglutide in adults with obesity and without T2D led to a ~15% mean body weight reduction over 68 weeks[117], with newer generation dual incretin agonists such as tirzepatide leading to even greater weight loss[118]. In a population with T2D and high rates of established CVD and CKD, semaglutide reduced major adverse cardiovascular events (MACEs) by 26% over a mean of 2.1 years[119]. In people with obesity and without T2D, a 20% reduction in MACEs and a 22% reduction in the composite kidney end point were observed over ~3.3 and ~3.5 years, respectively[120,121]. In another large trial, semaglutide reduced kidney failure or ≥50% eGFR decline by 21% in people with both T2D and CKD over ~3.1 years[122].

Despite their multisystem benefits, the uptake of GLP-1RAs remains limited, although prescriptions are rapidly increasing. In a commercially insured US T2D population, GLP-1RA uptake increased from 3.2% to 10.7% between 2015 and 2019, with lower uptake among Asian, Black, Hispanic, male and low-income patients[123]. International data show similarly low uptake in Australia, as only ~10% of people with T2D and CKD — and 11.3% of individuals meeting the GLP-1RA

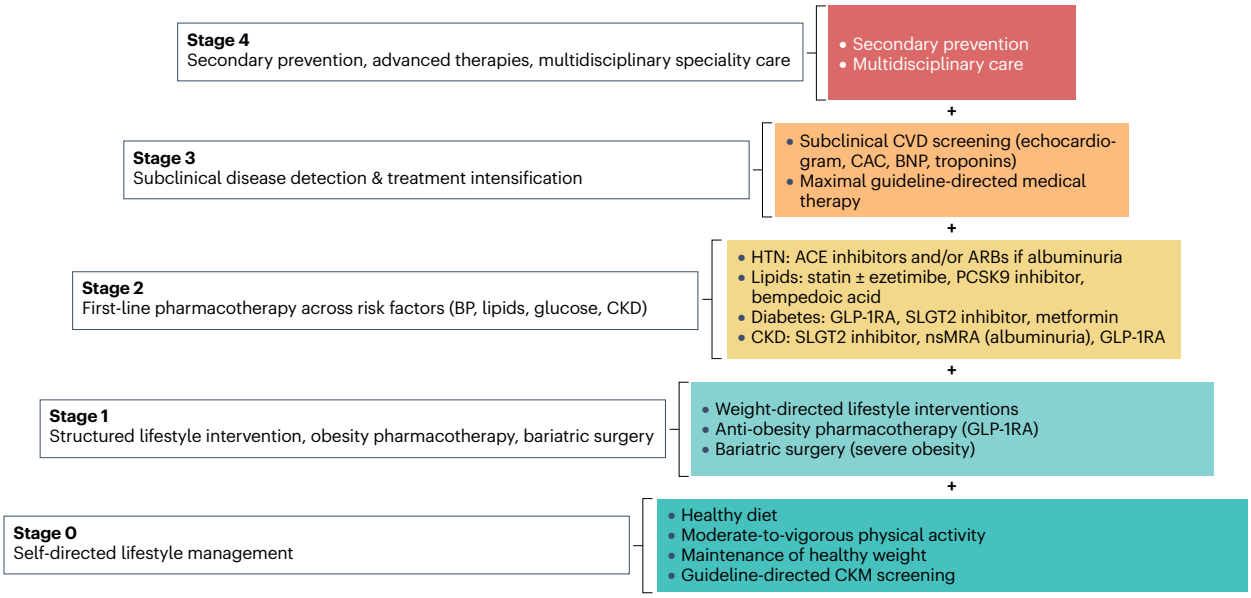


Fig. 5 | Key tenets of cardiovascular–kidney–metabolic syndrome management. Cardiovascular–kidney–metabolic (CKM) syndrome is managed along a continuum from Stage 0 (primordial prevention with healthy diet, physical activity, weight maintenance and guideline-directed screening) through to Stage 4 (secondary prevention, advanced therapies and multidisciplinary care). Interventions escalate from structured lifestyle and weight-directed therapies (Stage 1), to first-line pharmacotherapy across blood pressure, lipid, glycaemic and kidney domains (Stage 2), to intensified treatment guided by subclinical cardiovascular disease (CVD) and advanced chronic kidney disease (CKD) (Stage 3), and finally to advanced disease management (Stage 4). ACE, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; BP, blood pressure; BNP, brain natriuretic peptide; CAC, coronary artery calcium; GLP-1RA, glucagon-like peptide-1 receptor agonist; HF, heart failure; HTN, hypertension; nsMRA, non-steroidal mineralocorticoid receptor antagonist; PCSK9, proprotein convertase subtilisin/kexin type 9; SGLT2, sodium–glucose co-transporter 2.

kidney outcome trial inclusion criteria[122] – were prescribed a GLP-1RA in 2020–2021 (ref. 124). Later estimates show a near five-fold increase in GLP-1RA use in the USA between 2018 and 2023 (ref. 125); nonetheless, most eligible patients are not prescribed these drugs. Costs remain high and insurance coverage limited, particularly for people without T2D. Gastrointestinal side effects and a preference for oral over injectable medications might also drive drug discontinuation, although new oral formulations of these drugs might mitigate this barrier[126]. Of note, over-representation of patients at low CVD risk among GLP-1RA users might reflect gaps in prescriber education[127].

Sodium–glucose co-transporter 2 inhibitors

Large RCTs in people with T2D demonstrate that SGLT2 inhibitors improve glycaemic control, reduce HF with preserved ejection fraction and HF with reduced ejection fraction hospitalizations and deaths, slow CKD progression and modestly reduce MACEs in existing ASCVD[128–132]. HF and CKD benefits are similar in populations with and without T2D[131–133]. Modest improvements in early-stage CKM risk factors, including body weight and systolic blood pressure, are also well established[134].

Patterns and rates of SGLT2 inhibitor uptake mirror those reported for GLP-1RAs. In a broad US cohort from OLDW, between April 2022 and March 2023, only 11.4% of adults with T2D and a Class 1a recommendation (strongly recommended; high-quality evidence) for SGLT2 inhibitors were prescribed one[135]. Among adults without T2D and a Class 1a recommendation, the prescription rate was markedly lower, at 3.1%[135]. In a large US registry of ambulatory patients with heart failure, rates rose from 4.6% to 16.2% between 2019 and 2023[136]. SGLT2 inhibitor uptake was particularly low among older individuals, women, those without T2D, those with low educational attainment and income, and those at highest risk of CKM[127,135,136]. In other high-income countries, prescription rates are similarly low but rising[137–139]. For example, in a cross-sectional study of UK adults with CKD in 2022, only 17% of patients with a guideline-based indication for SGLT2 inhibitors were prescribed one[140]. Similar to GLP-1RAs, SGLT2 inhibitors are expensive, although US and European patent expirations might allow generic competition soon. Additionally, clinician comfort prescribing these drugs might be limited, particularly in patients at a high risk who might benefit the most[135,137]. Tolerability of SGLT2 inhibitors is generally better than that observed for GLP-1RAs[141], suggesting that barriers to long-term population-level impact might be more surmountable.

Non-steroidal mineralocorticoid receptor antagonists

nsMRAs are the newest clinically available therapy class targeting CKM, with finerenone being the first agent to gain regulatory approval. In large RCTs, finerenone added to standard of care significantly improved kidney and cardiac outcomes in people with CKD and T2D[142,143]. Benefits included an 18% relative risk reduction in CKD progression and a 13% relative risk reduction in MACEs. Finerenone also reduced MACEs in people with HF and mildly reduced or preserved ejection fraction[144]. These results have established nsMRAs as an effective pharmacological option for managing intermediate-to-late-stage CKM syndrome. Ongoing trials are assessing the efficacy of nsMRA use in combination

	GLP-1	SGLT2 inhibitor	nsMRA
MACE	↓ 21%	↓ 11%	↓ 15%
HF hospitalization	↓ 28%	↓ 30%	↓ 26%
CKD progression	↓ 21%	↓ 33%	↓ 17%
HbA_{1c}	↓ 2.1%*	↓ 1%**	Neutral
Body weight	↓ 8.6kg*	↓ 2kg	Neutral

Fig. 6 | Effects of novel therapeutics on cardiovascular–kidney–metabolic endpoints. All estimates derive from systematic review and meta-analyses[164–169]. Chronic kidney disease (CKD) progression is defined as a sustained ≥40 % or 50 % decrease in estimated glomerular filtration rate from baseline, doubling of baseline serum creatinine, kidney deaths or kidney failure. Major adverse cardiac events (MACEs) are defined as myocardial infarction, stroke or cardiovascular death. Green denotes a positive effect (that is an end-point reduction). *Refers to tirzepatide, the best-in-class glucagon-like peptide-1 receptor agonist (GLP-1RA) for glycaemic control and weight loss. **Refers to canagliflozin, the best-in-class sodium–glucose co-transporter 2 (SGLT2) inhibitor for glycaemic control. HbA_{1c}, glycated haemoglobin; HF, heart failure; nsMRA, non-steroidal mineralocorticoid receptor antagonist.

with SGLT2 inhibitors in heart and kidney disease based on promising subgroup data[145].

Despite these benefits, real-world uptake of nsMRAs remains extremely limited and large studies of usage are unavailable. Major clinical guidelines (KDIGO and the American Diabetes Association) endorsed finerenone in late 2023, but high costs, prior authorization requirements and concerns regarding adverse events such as hyperkalaemia have resulted in slow uptake[146].

Emerging therapeutics

Several therapeutic approaches might expand options for CKM syndrome management. Long-acting lipid-lowering therapies — including inclisiran, gene-editing approaches and RNA-based agents targeting lipoprotein(a) — are in development and might reduce cardiovascular risk[147–149]. Aldosterone synthase inhibitors (for example, lorundrostat) might offer cross-domain benefits similar to nsMRAs, with efficacy in resistant hypertension and reductions in albuminuria in CKD demonstrated in phase III and phase II clinical trials, respectively[150,151]. The incretin pipeline might hold the greatest promise for pleiotropic CKM benefit. Beyond inducing weight loss, oral GLP-1RAs have demonstrated cardiovascular efficacy in people with ASCVD and T2D in an RCT[152], and next-generation multi-agonists (retatrutide) and amylin-targeting combinations (cagrilintide plus semaglutide) have produced greater weight loss than currently approved incretin agonists in phase II trials[153,154]. Cardiovascular and kidney outcome trials are underway.

As with established therapies, the population-level impact of emerging agents will depend not only on clinical efficacy but also on overcoming the barriers of cost, access and equitable uptake that continue to limit the reach of proven CKM treatments.

Appraising the cardiovascular–kidney–metabolic syndrome framework

As a newly introduced paradigm, the CKM syndrome framework has received limited formal commentary from established kidney societies[155]. Ultimately, its utility will depend on whether it improves patient and provider awareness, disease monitoring, treatment and outcomes in a cost-effective manner. A balanced appraisal of the paradigm requires considering its potential benefits, as well as its limitations and criticisms raised.

Opportunities

A key aim of the CKM syndrome staging system is to promote early intervention[2]. By identifying individuals at stage 1 — before the onset of metabolic risk factors, CKD or CVD — the framework creates prevention opportunities when quality of life has not been substantially compromised and lifestyle interventions might be most effective and easier to implement. Although the stages themselves are only beginning to be incorporated into major society guidelines, the AHA provides several recommendations for their clinical use. To screen adults aged ≥21 years with no known disease (stage 0), BMI and waist circumference should be measured annually and testing for metabolic syndrome (MetS) components performed every 3–5 years. For adults with stage 1 CKM syndrome or a history of gestational diabetes, MetS screening should be increased to every 2–3 years. At stage 2, annual evaluation for MetS is recommended, along with measurement of urinary albumin-to-creatinine ratio and eGFR, with more frequent testing in people with a high KDIGO CKD risk. Official guidance on screening for other CKM syndrome complications (for example, MASLD, diabetic complications, peripheral artery disease and HF) has not yet been defined[2].

The CKM construct has also advanced CVD risk prediction. In 2023, the PREVENT equations were developed to estimate 10- and 30-year CVD risk in adults in the USA aged 30–79 years[156]. Unlike earlier models, such as the pooled cohort equations or Framingham, PREVENT incorporates markers of kidney health (eGFR, urinary albumin-to-creatinine ratio), MetS (HbA_{1c}) and a social deprivation index, reflecting the known risk multiplier associated with social determinants of health. Notably, PREVENT predicts HF risk in addition to the risk of ASCVD. The equations have demonstrated improved calibration and similar or better discrimination compared with the pooled cohort equations in diverse and contemporary US populations[157,158], addressing the tendency of prior tools to overpredict risk overall and misclassify ASCVD risk in women[159]. PREVENT was included in the 2025 multi-society hypertension guideline and the 2026 multi-society dyslipidaemia guideline in the USA[160,161], underscoring how the CKM framework has started to shape clinical practice. International validation is ongoing and will be crucial for global adoption.

The CKM framework provides a conceptual basis for understanding pleiotropic benefits of novel therapeutic classes. SGLT2 inhibitors, GLP-1RAs and nsMRAs were initially developed for specific indications but have since demonstrated benefits across CKM outcomes. By framing these conditions as manifestations of a shared syndrome, the CKM construct encourages clinicians to consider benefits more broadly rather than restricting focus to a single organ. This integrated perspective might accelerate adoption of guideline-directed therapies in patients who stand to benefit across multiple CKM domains.

Finally, the CKM framework might raise patient and provider awareness of the interconnectedness of cardiovascular, kidney and metabolic disease and promote integrated, multidisciplinary screening and care across primary care, cardiology, nephrology and endocrinology.

Concerns

Despite potential benefits, several criticisms of the CKM framework merit consideration. First, the classification system labels a large majority of the population as having disease. In high-income countries, an estimated 75–90% of adults meet criteria for stage 1–4 CKM syndrome[7,8,11,12]. Labelling such a large proportion of the population as diseased might increase anxiety, promote unnecessary interventions and escalate health care costs.

Second, the staging system implies a linear progression from excess adiposity (stage 1) through metabolic risk factors or CKD (stage 2) to subclinical CVD (stage 3) and clinical CVD (stage 4). This trajectory does not fully reflect the heterogeneity of CKM disease. Individuals can develop CVD without obesity or clear metabolic risk factors, bypass intermediate stages or present with advanced kidney disease without CVD[31]. Although outcomes worsen with advancing stage, the relationship is not uniformly graded, with stage 1 conferring little mortality risk over stage 0 in some cohorts[5,26].

Third, the CKM framework might be cardiocentric. Kidney failure — a devastating disease — is classified as stage 3 CKM unless accompanied by CVD. Notably, people with advanced kidney disease have been largely excluded from cardiovascular trials underpinning CKM-directed therapies[162]. The CKM framework incorporates only CVD risk prediction equations (that is, PREVENT) but not kidney failure prediction equations (for example, Kidney Failure Risk Equation)[163]. Moreover, primary kidney disorders with non-cardiometabolic causes (for example, glomerulonephritis and genetic kidney diseases) do not fit well in the CKM framework. Finally, despite close ties to metabolic dysfunction, MASLD has not been incorporated into the CKM syndrome staging system[2].

The real-world implications of the CKM framework for patient care remain unclear. Substantial barriers hinder the implementation of the sort of integrated, multidisciplinary care envisioned by the AHA as many health care systems remain siloed by specialty, and mechanisms for coordinated CKM care are underdeveloped. Moreover, the fundamental question remains as to whether the integrated CKM paradigm will prove more clinically or operationally useful than conceptualizing cardiovascular, kidney and metabolic disease separately.

Conclusions

CKM syndrome is a unifying construct capturing biological, social and environmental interactions linking cardiovascular, kidney and metabolic disease. The CKM syndrome staging system emphasizes risk stratification and early intervention. Epidemiological data show that most adults worldwide meet the criteria for the syndrome, with prevalence rising over recent decades and disproportionate burdens among disadvantaged racial, geographic and socio-economic groups. Although several therapeutics such as SGLT2 inhibitors, GLP-1RAs and nsMRAs offer multisystem benefits, real-world uptake remains constrained by cost, access and prescribing disparities. Emerging prediction tools, such as the PREVENT equations, underscore the potential of CKM syndrome as an integrated clinical decision-making framework. Combating CKM syndrome globally will require coordinated strategies combining biological insights, effective therapies, clinical innovation and policy solutions, while ensuring that therapeutic advances reach those at highest risk.

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